

8 *Use of the Data Booklet is relevant to this question.*

A student mixes 20.0 cm^3 of 5.00 mol dm^{-3} sulfuric acid with an equal volume of 6.00 mol dm^{-3} sodium hydroxide. The initial temperature of both solutions is 25.0°C . The maximum temperature reached after the reaction is 55.0°C . Assume the density of both solutions is 1 g cm^{-3} .

What is the value of the enthalpy change of neutralisation, in kJ mol^{-1} , calculated using these values?

- A -41.8
- B -50.2
- C -83.6
- D -100.3

Commented [SXF(5): Neutralisation and I included the units cos the A level qns has the units in the options